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Analysis of similar weather conditions to improve reuse in weather-based decision support systems

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ABSTRACT

Weather-based decision support systems (DSSs) are being built to improve the efficiency of the production systems in the domains of healthcare, agriculture, transport, governance and so on. Normally, a weather condition (WC) is represented by the statistical values of weather variables for a given duration (e.g. a day or a week). In a weather-based DSS, given a WC, the domain experts prepare the appropriate suggestions to improve the efficiency of the stakeholders. Normally, once the domain experts prepare a suggestion for a given WC that belongs to certain period (e.g. year or season), there is a scope to *reuse* the same suggestion for the similar WCs of other period(s). As a result, the performance of the DSS could be improved. In this paper, to improve *reuse*, we have proposed a notion of *category-based WC (CWC)* which is formed by using the categories of weather variables in the respective domain. By considering the context of agromet advisory service operated by India Meteorological Department (IMD) and the corresponding weather categories provided by IMD, we have analyzed the extent of *reuse* among CWCs by conducting the experiments on 30 years of weather data collected at Rajendranagar, Hyderabad, Telangana state. The experiments are conducted by considering two types of CWCs with the duration equal to one day and five days. By varying the number of weather variables in CWC from one to five, we have computed the extent of *reuse* among CWCs of different periods of the following period types: year, season, and phenophases (i.e., growth stages) of the Rice crop. The results show that there is a significant similarity among the CWCs of the given period and the CWCs of the preceding periods of each period type. For any domain including agriculture, the results provide an opportunity to improve the efficiency of weather-based agricultural DSSs by improving the *reuse* of the weather-based suggestions.

1. Introduction

Weather is the state of atmosphere as measured on the scale of hot or cold, wet or dry, calm or storm, and clear or cloudy (Strahler and Strahler, 1984; Monkhouse, 1965). Some of the weather variables are rainfall, maximum temperature, minimum temperature, maximum relative humidity, minimum relative humidity, cloud cover, wind speed, and wind direction. Many decisions in our day-to-day life are influenced by weather. The weather observation and forecasting systems have become vital to every country for improving the efficiency of several systems which deal with life and property. Over the years, the weather information and forewarning systems are gradually becoming powerful, domain-specific and location-specific. Also, proper understanding/application of observed and forecast weather information/alerts improves the efficiency of planning and execution of tasks in all walks of life, occupations, and services. Governments are investing huge budgets for developing advanced computer information systems to forecast weather, and domain specific weather-based decision support systems.

Decision Support System (DSS) is an interactive computer-based information system that supports decision making in various domains. Weather-based DSSs are being built to improve the efficiency of the production systems in the domains of healthcare, agriculture, transport, governance and so on. In the literature, the notion of *reuse* is being exploited to improve the efficiency of DSSs in domains like clinical services (Johnson et al., 2014) and software project management (Srivastava, 2004). In the agriculture domain, an effort has been made to propose the notion of *weather window* to improve *reuse* (Balasubramanian et al., 2014). It was proposed that the suggestions prepared for the dominant *weather window* could be reused for other

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similar *weather windows* which occur in future. So far, no effort has been made to analyze the potential of *reuse* in weather-based DSSs.

In this paper, we proposed a methodology to improve the *reuse* in weather-based DSSs. A weather condition (WC) is represented by the statistical values of weather variables for a given duration (e.g. a day or a week). In a weather-based DSS, given a WC, the domain experts prepare the appropriate suggestions to improve the efficiency of the stakeholders. Normally, once the domain experts prepare a suggestion for a given WC that belongs to certain period (e.g. year or season), there is a scope to reuse the same suggestion for similar WCs of other period (s). As a result, the performance of DSS could be improved by reducing the repetition of work and operational cost, and improving the quality of suggestions. The repetition of work can be reduced by reusing the prepared content for the given WC for the similar WC. The operational cost could be reduced as the system could be operated by less number of experts. The quality of suggestions could be improved as experienced experts can be involved to prepare a comprehensive suggestion for the given WC. It has been observed that, the notion of WC which contains the corresponding statistics of weather variables (real values) is not efficient to exploit the *reuse* potential of weather-based suggestions. To improve *reuse*, the proposed methodology exploits weather category information available in the respective domain and proposes a notion of *Category-based Weather Condition (CWC)*. For the given domain, the domain experts divide the range of each weather variable into different categories based on its influence on the respective domain. The notion of CWC improves the *reuse*, as it improves the extent to find similar CWCs among the CWCs of different periods of the given period type.

We have analyzed the extent of *reuse* under the proposed framework by considering the case study of agromet advisory service operated by India Meteorological Department (IMD). We have conducted the experiments on 30 years of weather data collected at Rajendranagar, Hyderabad, Telangana State, India, by using the categories of weather variables provided by IMD and considering two types of CWCs with the duration equal to one day and five days. By varying the number of weather variables in CWC from one to five, we have computed the extent of *reuse* among CWCs of different periods by considering the period types, year, season, and phenophases (i.e., growth stages) of the Rice crop. The results show that there is a significant degree of *reuse* among the CWCs over the periods of the following period types: Year, Seasons, phenophase of the Rice crop. For any domain including agriculture, the results provide an opportunity to improve the efficiency of weather-based agricultural DSSs by improving the *reuse* of the weather-based suggestions.

The research is carried out as a part of investigation to build a system for improving the efficiency of weather-based agromet advisories (Maini and Rathore, 2011). An effort has been made in Mamatha et al. (2014) to develop a framework for exploring the extent of similar WCs. In this paper, we have extended the framework proposed in Mamatha et al. (2014) by varying the number of variables in WC and conducted the additional experiments by considering the period types as phenophases of the Rice crop. Also, the presentation has been improved significantly.

The remaining part of the paper is organized as follows. In the next section, we present the related work. In Section 3, we explain about CWC and methodology to compute *reuse*. In Section 4, we present the case study and experiment results. The last section contains the summary and conclusions.

2. Related work

The notion of *reuse* plays a key role in minimizing the human effort and reducing the software development cost. Software development life cycle consists of reusable components from the design stage to deployment stage (Srivastava, 2004). The concept of software component reuse to build a new system is discussed in Selby (2005). The notion of *Software Product Line (SPL)* refers to software engineering methods, tools, techniques for creating a collection of similar software systems from a shared set of software assets using a common means of production. An approach for domain-oriented geographical SPLs is explained by considering a case study of marine ecology sub-domain (Buccella et al., 2013). Also, various characteristics of software reuse in SPL at project level, design level, implementation level and deployment level were analyzed.

The notion of *reuse* is a widely used strategy to improve the efficiency of DSSs. A *temporal abstraction* mechanism for knowledge *reuse* to assess the child growth is discussed (Kuilboer et al., 1993). Here, *temporal abstraction* is based on the temporal reasoning that provide interpretations on the given data. An effort have been made to build ICT-based clinical DSS by exploiting *reuse* of clinical knowledge (Stanescu and Filip, 2009). The clinical reminder system has been developed to manage the work-flow of the various stakeholders (doctors, lab technician, patients etc.) (Johnson et al., 2014).

The forecast of the climate, a few months before the start of the crop season helps the farmer in efficient crop planning and management decisions (Jones et al., 2000). The impact of temperature variability on Wheat crop yield is analyzed in Senthold et al. (2011). The impact of both temperature and precipitation on crop model predictions is presented in Riha et al. (1996). The process and operation of Agro-meteorological Advisory Service (AAS) by India Meteorological Department (IMD) is presented in Rathore (2013). In AAS, based on the weather forecast the agro-meteorological advice is provided twice a week with respect to all crops and livestock, which is disseminated to farmers and other stakeholders. The survey presented in Maini and Rathore (2011) show that farmers who followed the advisories provided under AAS had a net benefit of 10–15% in crop yield and 2–5% reduction in the cultivation cost as compared to farmers who have not followed the advisory provided under AAS. It was reported (Liu et al., 2015) that the temperature and precipitation values in the different seasons influence the travel mode selected by the travelers. The impact of temperature on human mortality rate is presented in Guo et al. (2011).

A prototype of an IT-based agro-meteorological system is proposed in Reddy et al. (2014). The system contains a mechanism to exploit the *reuse* of already prepared agromet advisory for a given weather condition, say *x*, for the preparation of agromet advice if the weather forecast is similar to *x*. A framework to *reuse* the content to improve the performance of an agro-advisory system is proposed in Mahadevaiah et al. (2012). In Balasubramanian et al. (2014), an approach was proposed to improve the efficiency of agro-meteorological advisory based on the concept of *dominant weather situations* that occurred over a period of two years by preparing the pairs of weather windows of the form <dominant weather condition, advisory>. It was reported that the concept of weather window has reduced the advice preparation overhead for new dominant weather situations. The importance of a web-based DSS for minimizing the climate risks in agriculture is discussed in Breuer et al. (2008). A framework is proposed in Krishna Reddy et al. (2012) to improve the practical agricultural education using the notion of *virtual crop labs*.

By considering the weather categories provided by IMD (Mazumdar and Medha, 2008), an effort has been made in Mamatha et al. (2014) to propose a framework to compute the extent of similarity among the Category-based Weather Conditions (CWCs) of a given *year* and *season* (Kharif, Rabi and Summer) and the preceding years and seasons. In this paper, we have expanded the methodology proposed in Mamatha et al. (2014) to compute the similarity of CWCs by varying the weather variables in *WC* from one to five. Furthermore, in addition to *year* and *seasons* (Kharif, Rabi and Summer), we have investigated the extent of similarity among the CWCs by considering the period types of different phenophases (i.e., growth stages) of the Rice crop.

None of the preceding works have analyzed the weather data to exploit *reuse* of weather related suggestions. The approach in this paper is different from the preceding approaches as we have proposed a

Table 1
Sample daily weather data from 1st March 2015 to 14 th March 2015 collected at Rajendranagar weather station, Hyderabad, Telangana state, India. The units of Tmax/Tmin, RHmax/RHmin and RF, are degree centigrade (°C), per cent (%) and millimeter (mm) respectively.

Date	Tmax	Tmin	RHmax	RHmin	RF
1-Mar-2015	33.0	17.0	85.0	58.0	0.0
2-Mar-2015	28.0	20.4	83.0	43.0	8.0
3-Mar-2015	31.0	19.5	75.0	34.0	0.0
4-Mar-2015	32.0	16.0	65.0	33.0	0.0
5-Mar-2015	32.0	18.2	61.0	44.0	0.0
6-Mar-2015	33.0	21.9	82.0	62.0	0.0
7-Mar-2015	32.0	19.5	91.0	44.0	13.0
8-Mar-2015	32.0	19.5	84.0	40.0	8.6
9-Mar-2015	32.0	19.0	75.0	63.0	0.0
10-Mar-2015	28.5	20.5	69.0	58.0	0.0
11-Mar-2015	28.5	16.0	52.0	26.0	0.0
12-Mar-2015	32.5	15.0	56.0	31.0	0.0
13-Mar-2015	32.5	16.0	82.0	21.0	0.0
14-Mar-2015	34.0	18.5	38.0	13.0	0.0

framework to analyze the extent of *reuse* by proposing the notion of CWC based on the weather category information available in the respective domain. Furthermore, by considering the case-study of AAS of IMD and by conducting the experiments on 30 years of weather data, we have demonstrated the potential of the proposed approach to improve the *reuse* in weather-based DSS.

3. Proposed approach

The weather of a given location is represented by the values of weather variables like rainfall (*RF*), maximum temperature (*Tmax*), minimum temperature (*Tmin*), maximum relative humidity (*RHmax*), minimum relative humidity (*RHmin*) and so on. The sample daily weather values for 14 days are given in Table 1. Apart from these five weather variables, there are other weather variables like fog, air pressure, so on. However, the proposed approach is a generic approach and can be extended to other weather variables.

The concepts of Weather Condition (WC) and Category-based Weather Condition (CWC) are the central ideas of the proposed approach.

The notion of WC represents the values of weather variables of a given location. Normally, the weather is represented for certain *duration* (d) which is number of days. If $d = 1$, the WC is equal to the collection of weather variable values of the given day. Otherwise, if $d > 1$, WC represents the statistics of weather variable values for d days.

Notably, weather data is collected in a continuous manner throughout the day. In this paper, we have proposed an approach by considering the weather data reported on a daily basis.

The notation of WC is defined as follows.

Definition 1 (*Weather condition (WC)***).** WC_i is defined as 6-ary values $\langle i, n, s, d, V, \psi(V) \rangle$. Here, i is the unique identifier, n ($n \geqslant 1$) represents the number of weather variables, s represents the start date, d represents the duration which is equal to the number of subsequent days including s, V represents the set of n weather variables, i.e., V = { $v_1, v_2, ..., v_n$ }, and $\psi(V)$ represents the set of statistical values of the corresponding weather variables in V over d i.e., $\psi(V)$ = { $\psi(v_1), \psi(v_2), ..., \psi(v_n)$ }. The notation $\psi(v_i)$ represents the statistics of v_i computed for d days using the function $\psi()$.

We use the notation $WC_i(n, d)$ to represent a WC with $\langle i, n, s, d, V, \psi(V) \rangle$.

Note that for the given v_i, appropriate $\psi()$ should be employed to compute the statistical value of v_i over d. The $\psi(v_i)$, where v_i = Tmax/Tmin/RHmax/RHmin is computed using the average of v_i over d whereas $\psi(RF)$ is computed as the sum of RF values over d (Mazumdar and Medha, 2008). Table 1 shows weather data from 1st March 2015 to 14th March 2015, collected at Rajendranagar weather station, Hyderabad, India. From the weather data in Table 1, we form the weather condition $WC_i(n, d)$ with unique id $i = 1$, $n = 5$, $d = 1$ as $WC_1(5, 1) = \langle$ 1, 5, 1-March-2015, 1, { Tmax, Tmin, RHmax, RHmin, RF }, { 33.0, 17.0, 85.0, 58.0, 0.0 } $\rangle$. Also, we form $WC_i(n, d)$ with unique id $i = 2$, $n = 5$, $d = 5$ as $WC_2(5, 5) = \langle$ 2, 5, 1-March-2015, 5, { Tmax, Tmin, RHmax, RHmin, RF }, { 31.2, 18.22, 73.8, 42.4, 8 } $\rangle$.

Given weather data, we are interested in computing the similarity among WCs. The criteria to compute the similarity of two WCs is as follows.

Definition 2 (*Similar weather conditions***).** Let $WC_i(n, d) = \langle i, n, s, d, V, \psi(V) \rangle$ and $WC_j(n, d) = \langle j, n, s, d, V, \psi(V) \rangle$. We say, $WC_i(n, d)$ and $WC_j(n, d)$ are similar if the values of n, d, V, and $\psi(V)$ for $WC_i(n, d)$ and $WC_j(n, d)$ are equal. Otherwise, we say $WC_i(n, d)$ and $WC_j(n, d)$ are dissimilar.

It can be observed that, normally, weather variable values are real numbers (Table 1). So, given two WCs, there is less possibility to get similar pairs of WCs using the preceding definition as two WCs are dissimilar even if one of the weather variable value differs by a small value. Notably, for a given domain, a small change in the value of weather variable may not affect the corresponding suggestions. For instance, in healthcare or agro-meteorology domain, a slight change in weather variable values like 0.2 degree centigrade in temperature value, or 2 per cent change in humidity values does not result into different patient care or crop management suggestions. Based on this observation, we propose to incorporate the notion of the *category* information of weather variable available for the respective domain to improve the similarity among WCs. Given the domain, it is assumed that the range of each weather variable is divided into different classes based on the influence on the application area or domain as determined by the domain specialists. Each class is termed as a category. For example, in the agriculture domain, domain specific categories of RH weather variable can be divided into *low*, *moderate*, *high* and *very high*. Consider relative humidity values for daily WC from Table 1, if the relative humidity range from 31 per cent to 60 per cent then, the relative humidity value is categorized as *moderate* (Mazumdar and Medha, 2008). Note that the categories and the corresponding duration of a weather variable may differ from domain to domain. For example, the convenient temperature range for the human beings is different than the convenient temperature range for crop and livestock.

We now explain the notion of weather category table (WCT) for a weather variable.

Weather Category Table (WCT): For weather variable v_i and the set C(v_i) of the corresponding weather categories, WCT(v_i) is a two column table which represents the mapping of each value of v_i to the corresponding category in C(v_i). Each row in WCT(v_i) is of the form < range of v_i, category name >. Based on the domain, the values of v_i are divided into different ranges and each range is assigned a category name. Note that, it is assumed that the WCT of weather variables exists for the given domain. The WCT is determined by the domain experts based on the influence of the weather variables in the respective domain.

We now define category-based WC based on WCT.

Definition 3 (*Category-based WC ($CWC_i(n, d)$)***).** Given n weather variables, WCT for n weather variables, $CWC_i(n, d)$ is $WC_i(n, d)$ with the following modifications. Each value of $\psi(v_i)$ in $WC_i(n, d)$ is replaced with C(v_i) as per $WCT(v_i)$. We represent $CWC_i(n, d)$ as $\langle$ i, n, s, d, V, C(V) $\rangle$. Here, C(V) = { $C(\psi(v_1)), C(\psi(v_2)), ..., C(\psi(v_n))$ }, where C($\psi(v_i)$) is the category value of $\psi(v_i)$ from WCT.

The criteria to compute similarity between two CWCs is as follows.

Definition 4 (*Similar CWCs***).** Let $CWC_i(n, d) = \langle$ i, n, s, d, V, C(V) $\rangle$ and $CWC_j(n, d) = \langle$ j, n, s, d, V, C(V) $\rangle$. We say, $CWC_i(n, d)$ and $CWC_j(n, d)$

are similar if the values of n, d, V, and C(V) are equal. Otherwise, we say $CWC_i(n, d)$ and $CWC_j(n, d)$ are dissimilar.

Normally, the weather phenomena repeats in cycles i.e., periodically. The granularity of cycle may depend on the requirement of the application. For example, we might be interested in finding how weather phenomena repeats each year, or winter season of each year. We define the notion of*period type (pt)* to denote the cycles of interest about weather phenomena. The *pt* depends on the application domain. We define the notion of *pt* as follows.

Definition 5 (*Period type (pt)***).** The period type (pt) is equal to the number of contiguous days. The notation pt_i represents the ith *pt*. The duration of pt_i is equal to the number of days in the corresponding *pt*.

Let pt_i be i^{th} *pt*. Given pt_i and weather data over D days (D ≫ duration of the pt), *k* periods = { p_1, p_2, ..., p_k } can be identified in D. For each period p_i, we compute CWCs of type CWC(n, d). The number of CWCs comes to $\lfloor D/d \rfloor$ (D ≫ d).

Normally, given a set of CWCs of p_i and p_j of the given pt_i, we are interested in computing the number of common CWCs among p_i and p_j. Suppose if we generate suggestions for CWCs of p_i, we are interested to know the number of suggestions which could be re-used for CWCs of p_j. To capture the extent of *reuse*, we define the metric *coveragepercentage* (*CP*) as below.

Definition 6 (*Coverage Percentage (CP) of a period p_i with respect to Pset (CP(p_i/Pset))***).** Let p_i and Pset (Pset ≠ ϕ) be a period and a set of periods of the given pt_i, respectively. Also, let $CWC(p_i)$ and *CWC(Pset)* be the set of CWCs of p_i and the set of CWCs of *Pset*, respectively. Consider x and y represent CWCs, and $x \in CWC(p_i)$ and $y \in CWC(Pset)$. We define the term *Cset* as the set of CWCs which are common in $CWC(p_i)$ and *CWC(PSet)*, i.e., *Cset* = {x/x ∈ CWC(p_i) and y ∈ CWC(Pset) and x&y are similar as per Definition 4}. The $CP(p_i/Pset)$ is equal to $\frac{|Cset|}{|CWC(p_i)|} \times 100$ i.e., equal to the percentage of CWCs of p_i in which p_i is similar to at least one CWC of *Pset*.

Given a sequence S = ⟨ p_1, p_2, ..., p_k ⟩ of *k* periods of period type pt_i computed from weather data *D* with *V* weather variables for *m* years, WCT(v_i) of each weather variable $v_i \in V$ and duration *d* of CWC, we present an approach to compute CP(p_i/*Pset*), for p_i = 2, ···, k where, *Pset* is the set periods of preceding to p_i. The pseudo code is provided in the Algorithm 1. We compute WCs of *k* periods as per the Definition 1 (refer Lines 1 to 3) and the corresponding CWCs as per the Definition 3 (refer Lines 4 to 6). Let *X* = CWCs of p_i and *Y* = CWCs of *k*-1 periods. The coverage percentage for p_i = 2, ···, k is computed as per the Definition 6 (refer Lines 8 to 11).

Algorithm 1. Compute CP(S, WCT, d)

Input : S: *k* periods of period type pt_i computed from the weather data of *m* years with *V* weather variables; WCT: Domain specific categories for *V* weather variables; *d*: duration of CWC.
Output: CP(p_i/Pset) for i = 2 , . . . , k.

1 **foreach** p_i *in S, where* i = 1 , . . . , *k,* **do**
2 | Compute WC(p_i) with *d* as per the Definition 1. /* WC(p_i) set of WCs for p_i */
3 **end**
4 **foreach** *WC*(p_i)*, where* i = 1 , . . . , *k,* **do**
5 | Compute CWC(p_i) using WCT as per the Definition 3. /* CWC(p_i) set of CWCs for p_i */
6 **end**
7 $Pset = p_1$
8 **foreach** p_i, *where* i = 2 , . . . , *k,* **do**
9 | Compute CP((p_i/*Pset*) as per the Definition 6.
10 | $Pset = Pset \cup p_i$
11 **end**

4. Case study

To demonstrate the effectiveness of CWC notion for exploiting *reuse*, we conduct the case study by considering the framework of Integrated Agromet Advisory Service (IAAS) (Maini and Rathore, 2011) of India Meteorological Department (IMD) (Mazumdar and Medha, 2008). We conducted experiments by considering the weather variable categories provided by IMD on 30 years (1986 - 2015) of weather data collected at the weather station of Rajendranagar, Hyderabad, Telangana state, India. In this section, we first explain about IAAS service of IMD and experimental setup by considering the context of IAAS. Next, we provide the experimental results.

4.1. Integrated agromet advisory service by IMD

Weather and climate information play a major role before and during the cropping season. The forecast of the weather and climate information provided in advance can help farmers in applying appropriate resources by taking advantage of favorable weather conditions and mitigate potential losses in unfavorable weather conditions. The agro-meteorological inputs improve agricultural production both in quantity and quality. In an environment of increasing weather and climate variability under climate change, farmers are in greater need of agro-meteorological information blended with weather sensitive management advisories before the start of the crop season and during the crop season to support the agricultural practices.

The India Meteorological Department (IMD) issues various ranges of weather forecast that helps in planning the appropriate measures for crop protection and management. IMD is currently providing forecasting services to farmers, fishermen, shipping, air navigation etc. IMD has started weather services for farmers from the year 1945. From 2008, IMD has started Integrated Agromet Advisory Service (IAAS) in collaboration with different organizations, for the duration of 5 days weather forecast. The main emphasis of IAAS system is to collect and organize climate/weather, soil, and crop information, and to integrate them with the weather forecast information to assist farmers in their crop management decisions. The agromet advisory is prepared based on the short and medium range weather forecasts. In IAAS, after receiving the weather forecast for the given region, the agromet experts prepare the agromet bulletin in local language based on the weather forecast, observed weather, crop stage and crop status. The agromet bulletin contains weather-based crop management steps for enhancing crop production to stabilize yields through the management of agro-climatic resources as well as other inputs such as water, fertilizer, and pesticides. The agromet advisory also contains the steps to enhance the livestock production. The agromet advisory bulletin will be disseminated to the farming community and related stakeholders.

Table 2
Categories of temperature, relative humidity and rainfall.

Weather variable name	Range	Category name
Temperature (°C) (Deviation from normal)	−1, 0, 1	Little change
	2 or −2	Rise/fall
	3–4	Appreciable rise
	−3 to −4	Appreciable fall
	5–6	Marked rise
	−5 to −6	Marked fall
	> 7	Large rise
	< = −7	Large fall
Relative humidity (%)	0–30	Low
	31–60	Moderate
	61–80	High
	> = 81	Very high
Rain fall (mm)	0–0	No rain
	0.1–2.4	Very light rain
	2.5–7.5	Light rain
	7.6–35.5	Moderate rain
	35.6–64.4	Rather heavy rain
	64.5–124.4	Heavy rain
	124.4–244.4	Very heavy rain
	> = 244.5	Extremely heavy rain

4.2. Experimental setup

The experiments are conducted by considering the weather variables, types of CWCs, categories in WCT, and period types based on the context of IAAS.

4.2.1. Weather variables

For the experiments, we consider the weather variables *Tmax*, *Tmin*, *RHmax*, *RHmin* and *RF*, which are being considered for preparing agromet advisories under IAAS. We have considered the following combinations of CWCs: { Tmax }, { Tmax, Tmin }, { Tmax, Tmin, RHmax }, { Tmax, Tmin, RHmax, RHmin }, { Tmax, Tmin, RHmax, RHmin, RF }.

4.2.2. Types of CWCs

The duration values of CWCs are fixed based on the types of weather forecast by IMD. Notably, IMD (Mazumdar and Medha, 2008) is issuing the following types of weather forecast based on the spatial and temporal scales of atmospheric systems and durations.

- Now Casting: The duration of the forecast is for the current day with few hours forecast ahead.
- Short Range Forecast: The duration of the forecast is 1–3 days.
- Medium Range Forecast: The duration of the forecast is 4–10 days.
- Long Range Forecast: The duration of the forecast is 10 days to a season (3–4 months).

For the experiments, we have considered the duration values of CWCs as one day ($d = 1$) and five days ($d = 5$). The CWCs with $d = 1$ are considered to analyze the extent of *reuse* for *Now Casting* and *Short Range Forecast*, and the CWCs with $d = 5$ are considered to analyze the extent of *reuse* for *Medium Range Forecast*.

4.2.3. Categories of weather variables

For preparing weather-based summaries and agromet advisories, IMD has defined categories for the weather variables temperature, humidity and rainfall (Mazumdar and Medha, 2008). The different ranges of each weather variable and the corresponding categories are given in Table 2. The category details of temperature, relative humidity and rainfall are as follows.

- **Categories of temperature:** The categories of *temperature* (Tmax or Tmin) are as follows: Little change, Rise/Fall, Appreciable rise, Appreciable fall, Marked rise, Marked fall, Large rise and Large fall. The categories of temperature are assigned by comparing the temperature value with its historical normal. In Table 2, the range of temperature categories indicates the deviation from the historical normal. Regarding the historical normal (Mazumdar and Medha, 2008), the climatologists define the historical normal as the arithmetic mean of a climate variable for 30-year interval. Climatologists understand the trends of both forecast and observed data using the corresponding historical normals.
- **Categories of relative humidity:** The categories of *relative humidity (RH)* are as follows: Low, Moderate, High and Very high. The RH categories are assigned by dividing the range of RH as shown in Table 2.
- **Categories of rainfall:** The categories of rainfall (RF) are as follows: No rain, Very light rain, Light rain, Moderate rain, Rather heavy rain, Heavy rain, Very heavy rain and Extremely heavy rain. The RF categories are assigned by dividing the range of RF as shown in Table 2.

Table 3
Durations of Kharif, Rabi, Summer pts.

Period type (pt)	Sowing month	Harvesting month
Kharif	June	October
Rabi	November	March
Summer	March	May

4.2.4. Period types

We have considered three kinds of period types (pts).

- **pt = Year:** We consider the *pt* as number of days in a year. By considering *pt* as year, we analyze the extent of *reuse* over the years.
- **pt = Kharif season/Rabi season/Summer season:** We consider the *pt* as number of days in a crop season. In India, three crop seasons namely, Kharif, Rabi and Summer are defined based on the distinct weather situation on a longer term. By considering *pt* as a crop season, say *x*, we analyze the extent of *reuse* over the crop seasons of type *x* across the years. The names of the crop seasons and the corresponding durations are as follows. The corresponding details are shown in Table 3.
  - Kharif season: The duration of the Kharif season is from June to October. This is also called as *Rainy* season. The RF values during this season are higher than the other two seasons.
  - Rabi season: The duration of the Rabi season is from November of the given year to March of the subsequent year. This is also called as *Winter* season. The temperature values in this season are lower than the other two seasons.
  - Summer season: The duration of the Summer season is from March to May. Hot conditions prevail during this season.
- **pts = Phenophases of the Rice crop:** We consider *pt* as number of days in a *phenophase* of the Rice crop. The Rice crop duration is divided into several phenonphases and each phenonphase indicates

Table 4
Duration of pts (phenophases of the Rice crop) for Kharif and Rabi Seasons.

Period type (pt)	Kharif Season Starting Week #- Ending Week # (# of Weeks)	Rabi Season Starting Week #-Ending Week # (# of Weeks)
Sowing	W27 - W32 (6)	W50 - W1 (4)
Transplanting	W32 - W36 (5)	W1 - W4 (4)
Maxium tillering	W35 - W38 (4)	W5 - W9 ((5)
Panicle initiation	W39 - W44 (6)	W9 - W12 (4)
Flowering	W43 -W48 (6))	W12 - W16 (5)
Maturity	W48 - W52 (4)	W16 - W20 (5)

Table 5
Experimental settings.

Variable name	Values
Dataset	30 years of daily weather data from the year 1986 to the year 2015 with weather variables V = {Tmax, Tmin, RHmax, RHmin, RF}
Period Type (pt)	Year, Seasons (Kharif season, Rabi season, Summer season) (refer to Table 3), Phenophases (Sowing, Transplanting, Maximum Tillering, Panicle Initiation, Flowering, Maturity) of the Rice crop for Kharif season and Rabi season (refer to Table 4).
Type of WCs WC(n, d)	WC(1, 1) and WC(1, 5) with V = {Tmax} WC(2, 1) and WC(2, 5) with V = {Tmax, Tmin} WC(3, 1) and WC(3, 5) with V = {Tmax, Tmin, RHmax} WC(4, 1) and WC(4, 5) with V = {Tmax, Tmin, RHmax, RHmin} WC(5, 1) and WC(5, 5) with V = {Tmax, Tmin, RHmax, RHmin, RF}
Type of CWCs CWC(n, d)	CWC(1, 1) and CWC(1, 5) with V = {Tmax} CWC(2, 1) and CWC(2, 5) with V = {Tmax, Tmin} CWC(3, 1) and CWC(3, 5) with V = {Tmax, Tmin, RHmax} CWC(4, 1) and CWC(4, 5) with V = {Tmax, Tmin, RHmax, RHmin} CWC(5, 1) and CWC(5, 5) with V = {Tmax, Tmin, RHmax, RHmin, RF}

the stage of development in the life cycle of the Rice crop in relation to the climatic conditions. By considering *pt* as a *phenophase*, say *z*, we analyze the extent of *reuse* over the phenophases of type *z* across the years. Phenophase is defined as an observable stage or phase in the annual life cycle of a plant or an animal which is defined by the start date and the end date. The phenophases generally have a duration of a few days or a few weeks (Crimmins et al., 2015). The climatic requirement of each phase may be different and also each phase is sensitive to the certain environmental variables as well as management practices. For the given location, appropriate favorable conditions with respect to each phenophase should prevail to obtain good yield (Saran et al., 1989). The duration of the Rice crop is divided into the following phenophases: Sowing, Transplanting, Maximum Tillering, Panicle Initiation, Flowering, and Maturity. The duration (starting week - ending week) of the phenophases of the Rice crop for Kharif and Rabi seasons are shown in Table 4. The notation *Wi* in Table 4 indicates the standard meteorological week ($W1$ indicates the first week of January and $W52$ indicates the last week of December). It can be noted that the durations of the pheophase consists of weeks from two consecutive years. For instance, the duration of the *sowing* phenophase in Rabi season is from $W50$ to $W1$ i.e., 50th week of the given year to 1st week of the next year.

4.3. Experimental results

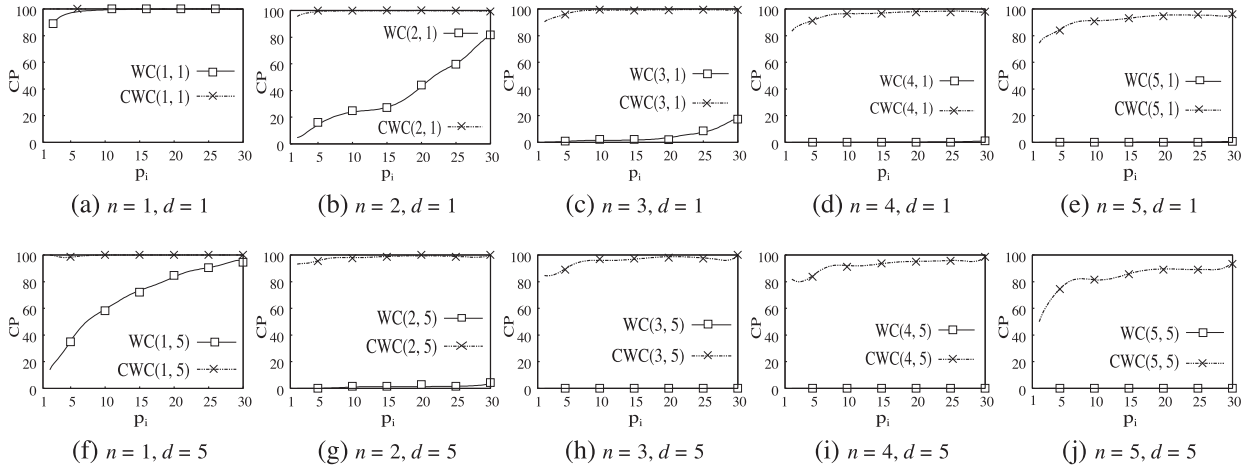
The details of the experimental settings are provided in Table 5. The dataset consists of 30 years of weather data for the weather variables *Tmax*, *Tmin*, *RHmax*, *RHmin* and *RF*, from 1986 to 2015, collected at Rajendranagar, Hyderabad, Telangana state. We have considered the following *pts*: Year, Kharif season, Rabi season, Summer season and the Phenophases of the Rice crop for Kharif season and Rabi season. The details of WCs and CWCs are as follows. By varying *n* values from 1 to 5 and *d* values as 1 and 5, we have considered ten types of WC(n, d): five types of WCs with *d* as 1 (WC(1, 1), WC(2, 1), WC(3, 1), WC(4, 1), WC(5, 1)) and five types of WCs with *d* as 5 (WC(1, 5), WC(2, 5), WC(3, 5), WC(4, 5), WC(5, 5)). Furthermore, by varying *n* values from 1 to 5 and *d* values as 1 and 5, we have considered ten types of CWC(n, d): five types of CWCs with *d* as 1 (CWC(1, 1), CWC(2, 1), CWC(3, 1), CWC(4, 1), CWC(5, 1)) and five types of CWCs with *d* as 5 (CWC(1, 5), CWC(2, 5), CWC(3, 5), CWC(4, 5), CWC(5, 5)).

The set of weather variables *V* for the WCs and CWCs are shown in Table 5. Given five weather variables ($n = 5$), several combinations could be formed for $n = 1$ to 4 and one combination is formed for $n = 5$. For $n = 1$ to 4, we have considered only one combination of weather variables for the formation of WC(n, d) and CWC(n, d) as the results for the other combinations are found to be similar. We have not reported the other results due to space constraints.

For a given *pt*, WC and CWC, we plot the ith period number p_i on the X-axis and the corresponding value of coverage percentage $CP(p_i/Pset)$ on the Y-axis, where *Pset* represents the preceding periods of p_i. The graphs show the regression lines of the results.

4.3.1. Experiments with pt = year

By considering $pt = year$, $n = 1$ to 5 and $d = 1$ and 5, Fig. 1 shows CP of WC(n, d) and CWC(n, d). The total number of periods for



(a) $n = 1, d = 1$ (b) $n = 2, d = 1$ (c) $n = 3, d = 1$ (d) $n = 4, d = 1$ (e) $n = 5, d = 1$

(f) $n = 1, d = 5$ (g) $n = 2, d = 5$ (h) $n = 3, d = 5$ (i) $n = 4, d = 5$ (j) $n = 5, d = 5$

Fig. 1. CP for pt = year.

pt = year for 30 year weather data comes to 30. The CP(p_i/Pset) values are plotted from p_i = 2 to p_i = 30. For instance, CP(4/3) means p_i = 4, and *Pset* = { p_1, p_2, p_3 } i.e., the set of periods preceding to p_4.

The observations regarding the performance of WC(*n*, 1), where *n* = 1 to 5 shown in Fig. 1a–e respectively, are as follows. It can be observed that the CP values for WC(1, 1) in Fig. 1a is 98% and for WC(2, 1) in Fig. 1b is 20% for the fifth period. Furthermore, it can be observed that the CP values for WC(3, 1), WC(4, 1) and WC(5, 1) in Fig. 1c, d, and e equals to 0 for the fifth period respectively. The reason for the reduction in the CP performance is as follows. Notably, the similarity between a pair of WCs (Definition 2) is computed by equating the real values of the corresponding weather variables. Normally, each value of weather variable receives a distinct real value, and as a result, the probability of exact match between two real values is further reduced. Also, as *n* increases, the probability of exact match is further reduced. As a result, the CP performance of WC(n, 1) is reduced significantly and reached to 0 as *n* increases from 1 (Fig. 1a) to 5 (Fig. 1e).

The observations regarding the performance of CWC(*n*, 1), where *n* = 1–5 shown in Fig. 1a–e respectively, are as follows. It can be observed that the CP values for CWC(n, 1) with *n* = 1–5, in Fig. 1a–e equals to 100%, 100%, 98%, 90%, and 81% for the fifth period respectively. It can be noted from each of Fig. 1a–e that as period number p_i increases, CP is increased significantly. Notably, the CP values are significantly high even at small p_i. For example, CP of CWC(3, 1) for the third period is 90%. This indicates that if we prepare the content/suggestions for the CWCs in a given domain for the first two years, 90% of the content/suggestions could be reused in the third year. Furthermore, it can be observed that CP of CWC reduces slightly as number of weather variables *n* increases from Fig. 1a–e. As expected, the probability of finding the match of CWC of p_i with the CWCs of *Pset* decreases with the increase in the value of *n*. Notably, the CP value has not reduced significantly as *n* increases. Based on these results, for *pt* = year, we can conclude that CP is increasing due to the notion of category as it is enabling more pairs of CWCs to become similar.

The observations regarding the performance of CWC(*n*, 5), where *n* = 1–5 shown in Fig. 1f–j respectively, are as follows. It can be observed that CP of WC(1, 5), WC(2, 5), WC(3, 5), WC(4, 5) and WC(5, 5) exhibits less performance as compared to that of the preceding experiments (with *d* = 1). Notably, the CP values of CWC(1, 5), CWC(2, 5), CWC(3, 5), CWC(4, 5) and CWC(5, 5) have reduced slightly as compared to the CP values of the corresponding preceding experiments with *d* = 1. This is due to that fact that the number of CWCs formed with *d* = 5 are less compared to the number of CWCs formed with *d* = 1. However, it can be observed that the CP performance of CWC(n, d) with *d* = 5 is significantly high even at the lower values of period number. For instance, it can be observed that the CP value of CWC(3, 5) equals to 90% in the fifth period.

Overall, we can conclude the following from the results of Fig. 1. Firstly, the notion of CWC provides a scope to improve the *reuse* as compared to the notion of WC. Secondly, significantly higher CP values are achieved with CWC(n, d) for *n* = 1–5 and *d* = 1 and 5 for *pt* = year even at lower period numbers.

It can be noted that the performance of WC is significantly low. Therefore, we will not report the results concerned to WC(n, d) in the remaining experiments.

4.3.2. Experiments with pt = Kharif season/Rabi season/Summer season

By considering the *pt* = Kharif/Rabi/Summer, *n* = 1–5, and *d* = 1 and 5, Fig. 2 shows CP of CWC(n, d). The total number of periods for 30 years weather data for Kharif season and Summer season comes to 30 and for Rabi season comes to 29.

The observations regarding the performance of CWC(*n*, 1), where *n* = 1–5 shown in Fig. 2a–e respectively, are as follows. From Fig. 2a, it can be noted that the CP performance of CWC(1, 1) for all three pts reaches 99% in the fifth period. It can be observed from Fig. 2b that the CP performance of CWC(2, 1) reaches 94% in the fifth period. Furthermore, it can be observed from Fig. 2c, d and e that the CP performance of three pts for CWC(3, 1), CWC(4, 1) and CWC(5, 1) by the end of the fifth period is about 90%, 80% and 78% respectively. It can be observed that, similar to the results of *pt* = year in Fig. 1, the CP performance of CWC reduces slightly as the value of *n* increases. However, the performance of CP is significantly high for higher values of p_i in Fig. 2a–e. For example, the CP for CWC(3, 1) for all three pts at p_i = 3 is about 80%. This indicates that if we prepare the suggestions/content for the CWCs of the first two periods, we can *reuse* 80% of the suggestions/content for the CWCs of the third period.

The CP of CWC(n, 5), where *n* = 1–5 for pt = Kharif/Rabi/Summer seasons are presented in Fig. 2f–j respectively. Notably, the CP values of CWC(1, 5), CWC(2, 5), CWC(3, 5), CWC(4, 5) and CWC(5, 5) have reduced as compared to the CP values of the preceding experiments with *d* = 1 for pt = Kharif/Rabi/Summer seasons. This is due to that fact that the number of CWCs for pt = Kharif/Rabi/Summer seasons formed with *d* = 5 are less compared to the number of CWCs formed in the corresponding pts with *d* = 1. However, it can be observed that the CP performance of CWC(n, d) with *d* = 5 for pt = Kharif/Rabi/Summer seasons is significantly high even at the lower values of period number. For instance, it can be observed that the CP value of CWC(3, 5) is about 80% in the fifth period.

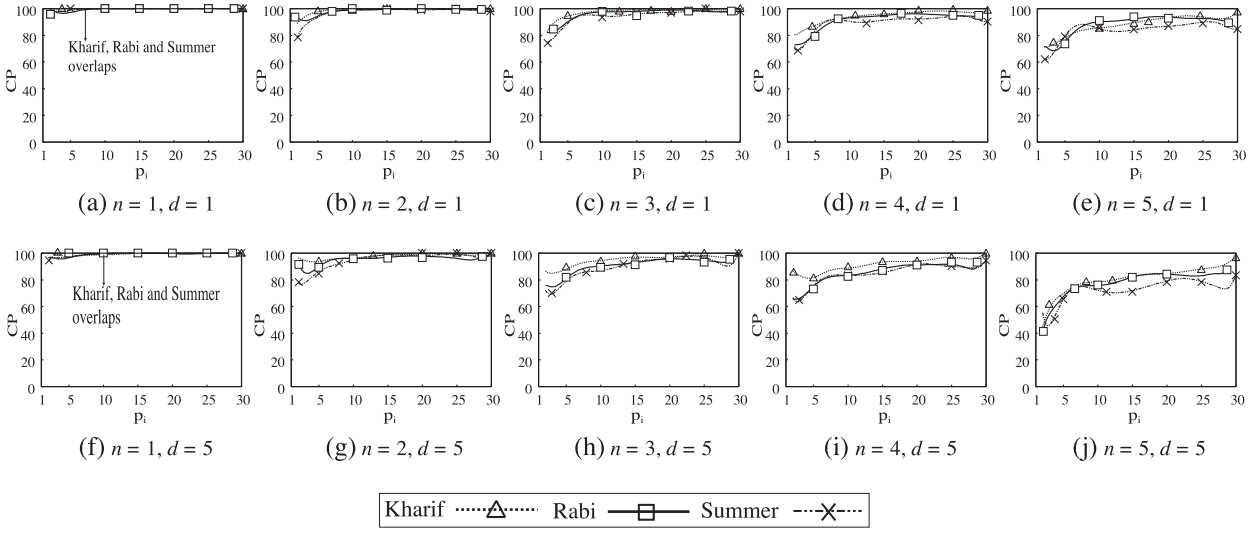


Fig. 2. CP for *pt* = *Kharif*/*Rabi*/*Summer seasons*.

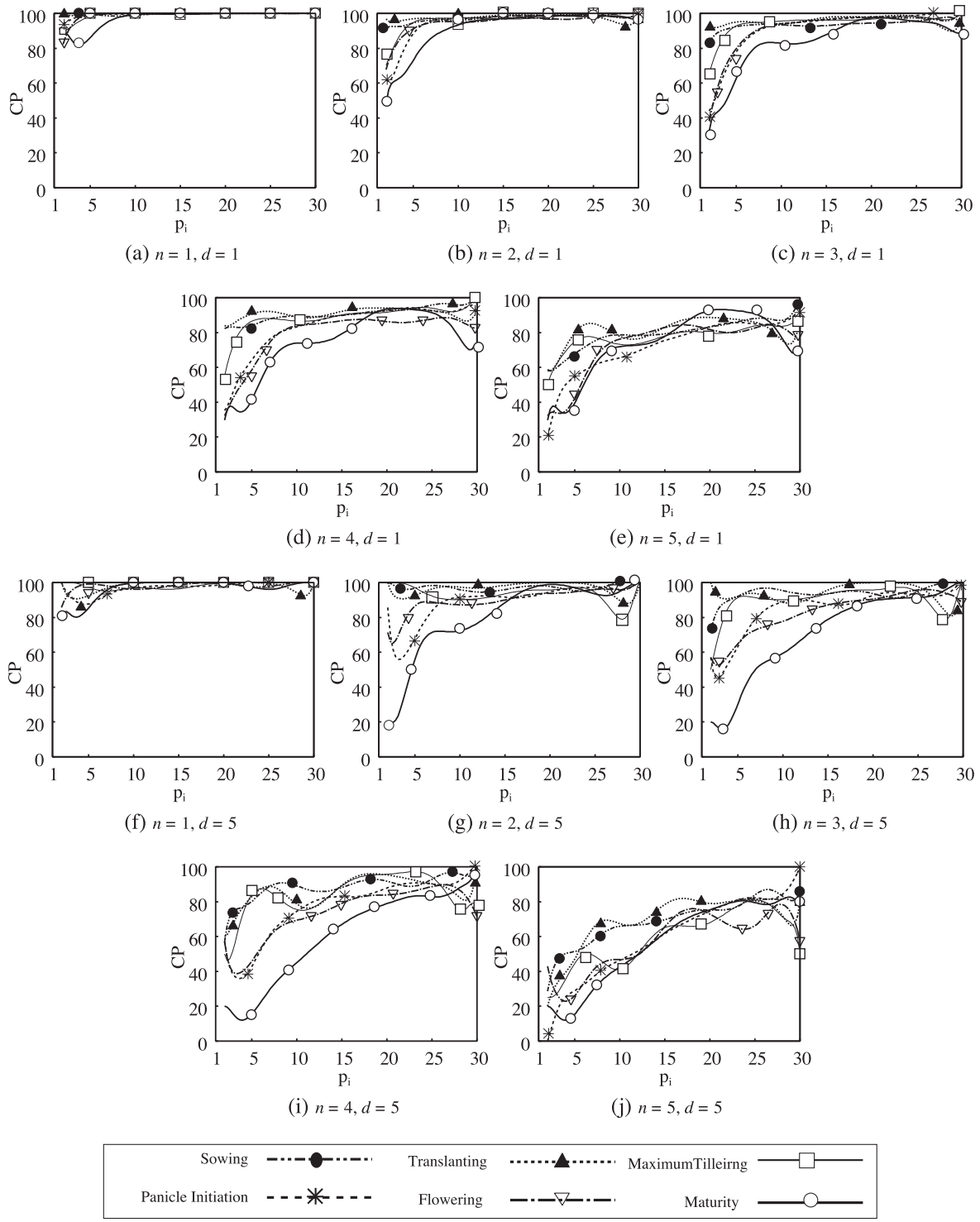


Fig. 3. CP for *pt* = Sowing/Transplanting/Maximum Tillering/Panicle Initiation/Flowering/Maturity (*Phenophase durations* of the Rice crop during the *Kharif season*).

We can conclude the following from Fig. 2. Firstly, the CP performance for $d = 1$ and 5 with respect to Kharif, Rabi and Summer seasons pts are found to be encouraging. Secondly, even though there is a slight decrease in the CP values in the experiments with $d = 5$ (Fig. 2f–j) as compared to the CP values in the experiment with $d = 1$ (Fig. 2a–e), the results of both experiments exhibit significantly higher value of CP as p_i increases. For example, the CP performance of CWC(3, 5) by the end of the third period reaches 80%.

Notably, it can be observed that the results for Kharif/Rabi/Summer seasons pts exhibit significantly higher values of CP (80% of CP at $p_i = 3$ for CWC(3, 5)) even though the duration of Kharif/Rabi/Summer seasons pts is equal to one third of the duration of the year pt. These results can be used to develop DSSs for the class of application domains in which CWCs of Kharif, Rabi and Summer influence the decisions.

4.3.3. Experiments with pts = Phenophases of the Rice crop

The names of phenophases (or *pts*) of the Rice crop are as follows: Sowing (S), Transplanting (T), Maximum Tillering (MT), Panicle

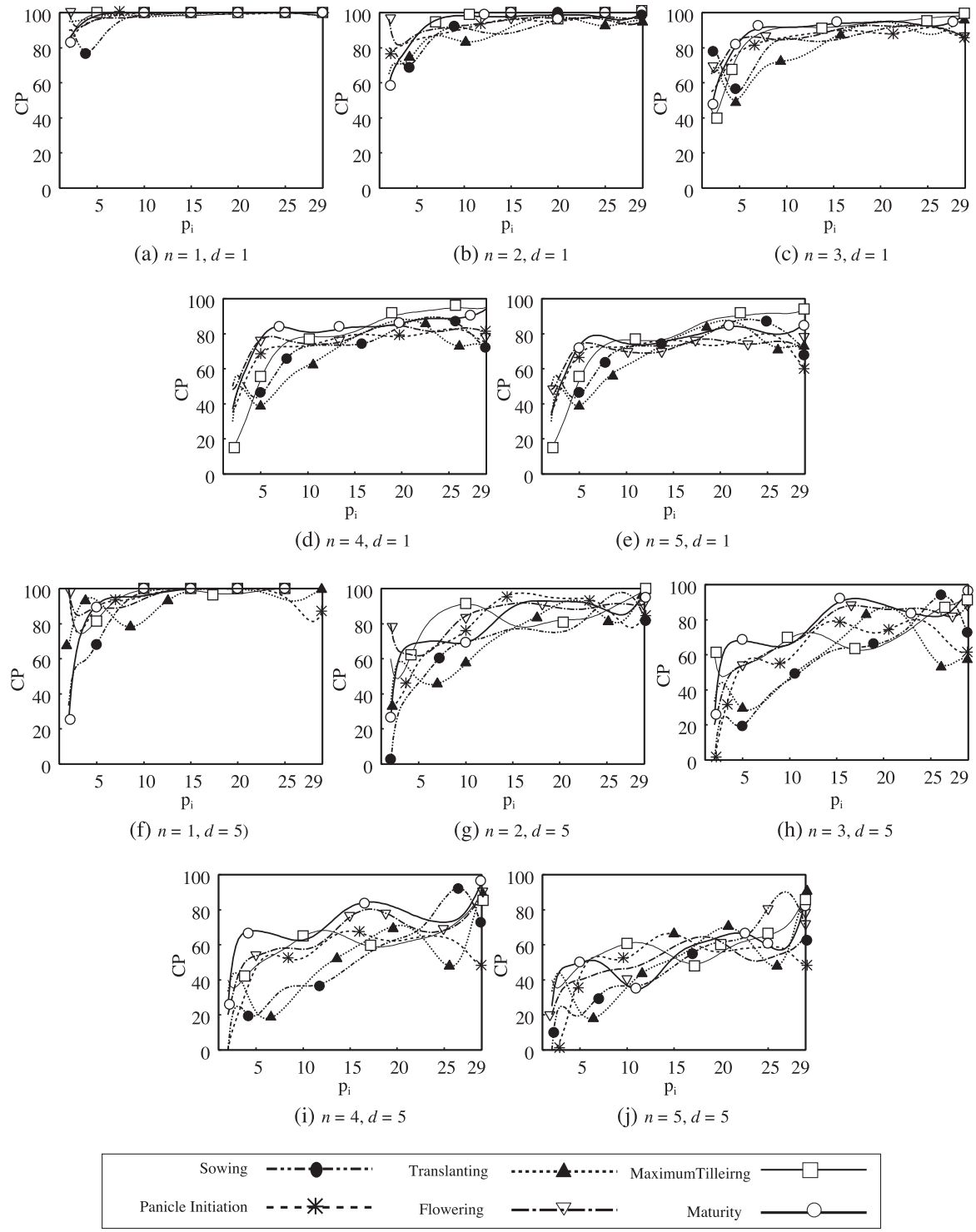


Fig. 4. CP for *pt* = Sowing/Transplanting/Maximum Tillering/Panicle Initiation/Flowering/Maturity (*Phenophase durations* of the Rice crop during the *Rabi season*).

Initiation (PI), Flowering (F), and Maturity (M). The corresponding durations of phenophases for Kharif and Rabi season are given in Table 4. Fig. 3 shows CP of CWC(n, d), where n = 1–5, and d = 1 and 5, for the pt = S/T/MT/PI/F/M phenophase of the Rice crop during the Kharif season. The total number of periods of each pt for 30 years of weather data for Kharif season comes to 30.

The observations regarding the performance of CWC(n, 1), where n = 1–5 shown in Fig. 3a–e respectively, are as follows. From Fig. 3a, it can be noted that the CP performance of CWC(1, 1) for all six pts reaches 80% in the fifth period. It can be observed from Fig. 3b that the CP performance of CWC(2, 1) reaches 75% in the fifth period. Furthermore, it can be observed from Fig. 3c, d and e that the CP performance of all pts for CWC(3, 1), CWC(4, 1) and CWC(5, 1) by the end of the fifth period is about 50%, 40% and 40% respectively. It can be observed that, similar to the results of Fig. 1 (pt = year) and Fig. 2 (pt = Kharif/Rabi/Summer), the CP performance of CWC reduces as the value of n increases. It can be observed that, the performance of CP is considerably high for higher values of p_i in Fig. 3a–e. For example, the CP for CWC(3, 1) for all pts at p_i = 5 is about 50%. This indicates that if we prepare the suggestions/content for the CWCs of the first four

Table 6
CP values for pt = year.

CWC (n, d)/period	CWC (1, 1)	CWC (2, 1)	CWC (3, 1)	CWC (4, 1)	CWC (5, 1)	CWC (1, 5)	CWC (2, 5)	CWC (3, 5)	CWC (4, 5)	CWC (5, 5)
3	100	98	**94**	82	80	100	94	**83**	82	60
5	100	99	**96**	87	81	98	96	**84**	87	76
7	100	100	**99**	98	90	100	97	**95**	90	81
10	100	99	**99**	96	88	100	97	**96**	91	81

Table 7
CP values for pt = Rabi season.

CWC (n, d)/period	CWC (1, 1)	CWC (2, 1)	CWC (3, 1)	CWC (4, 1)	CWC (5, 1)	CWC (1, 5)	CWC (2, 5)	CWC (3, 5)	CWC (4, 5)	CWC (5, 5)
3	100	90	**82**	74	70	95	82	**76**	62	50
5	100	95	**90**	80	74	97	85	**78**	70	60
7	100	98	**90**	90	88	98	94	**88**	81	78
10	100	100	**100**	97	97	100	94	**90**	81	78

periods, we can *reuse* 50% of the suggestions/content for the CWCs of the fifth period.

Fig. 3f–j shows the CP of CWC(n, 5), where n = 1–5, for pt = S/T/MT/PI/F/M phenophase of the Rice crop during the Kharif season. Notably, the CP values of CWC(1, 5), CWC(2, 5), CWC(3, 5), CWC(4, 5) and CWC(5, 5) have reduced as compared to the CP values of the preceding experiments with d = 1. This is due to the fact that the number of CWCs formed with d = 5 for pt = S/T/MT/PI/F/M are less compared to the number of CWCs of corresponding pt = S/T/MT/PI/F/M formed with d = 1. However, it can be observed that the CP performance of CWC(n, d) with d = 5 is high even at the lower values of period number. For instance, it can be observed that the CP value of CWC(3, 5) for the seventh period is about 70% for all pts except for the pt = *maturity*.

Fig. 4 shows CP of CWC(n, d), where n = 1–5, and d = 1 and 5, for the pt = S/T/MT/PI/F/M phenophase of the Rice crop during the Rabi season. The total number of periods for 30 years weather data for Rabi season comes to 29. It can be observed that the results of Fig. 4a–j are similar to the corresponding results in Fig. 3a–j respectively.

We can conclude the following from the results of Figs. 3 and 4. Firstly, the CP performance for d = 1 and 5 with respect to pts as phenophases of Kharif and Rabi seasons are found to be encouraging. Secondly, even though there is a slight decrease in the CP values in the experiments with d = 5 (Figs. 3f–j and 4f–j) as compared to the CP values in the experiment with d = 1 (Fig. 3a–e and Fig. 4a–e), the results of both experiments exhibit significantly higher values of CP as p_i increases.

Notably, as compared to the results for year pt in Fig. 1 and Kharif/Rabi/Summer seasons pts in Fig. 2, the results in Fig. 3 also exhibit relatively high values of CP (80% CP at p_i = 3 for CWC(3, 1)), even though the durations of pts (S/T/MT/PI/F/M) is significantly less (few weeks) as compared to the duration of year pt (52 weeks), and the duration of Kharif (22 weeks), Rabi (22 weeks) and Summer (13 weeks) pts. For instance, the CP performance of CWC(3, 1) reaches above 70% after five years and CWC(3, 5) is about 62% for all pts except for pt = maturity. It can be observed that the CP values improved with p_i. This shows an interesting observation that there is a high probability of recurrence of CWCs with respect to pts of phenophases of the Rice crop over the years. The results also show that there is a scope to improve the *reuse* performance of weather-based DSSs to deliver agricultural suggestions for the Rice crop.

4.3.4. Results summary

In the experiments, we have reported the results by considering five weather variables. In this section, we explain the observations about the effect of number of weather variables in CWC on CP performance. For ease of understanding, we show the CP values of CWC(n, d), where n = 1–5, d = 1 and 5, in Tables 6–9 for p_i = 3, 5, 7 and 10 based on the experimental logs. Tables 6–9 show the results of CP of CWC(n, d) for pt = year/Rabi season/Flowering phenophase(Kharif Season)/Flowering phenophase (Rabi Season) respectively. It can be observed that given p_i, the CP decreases with the increase in number of weather variables n in both CWC(n, 1) and CWC(n, 5). Also, it can be observed in Tables 6–9 that given CWC(n, d), where n = 1–5, d = 1 or 5, the CP value increases with p_i.

Normally, given a domain, all weather variables may not influence the decisions at a given time (Mousanejad et al., 2009). The experiments are conducted by considering weather data collected at Rajendranagar, Hyderabad, Telangana state . In this region, RF influences weather during rainy season (June-October), Tmin influences weather during winter season (November-March) and Tmax influences weather during summer season (March-May). Based on this observation, we analyze the performance of CWCs with three variables (highlighted in Tables 6–9). The results show that the CP performance of both CWC(3, 1) and CWC(3, 5) is very encouraging for all pts. The CP performance of CWC(3, 1) and CWC(3, 5) with pt = year for p_i = 3 is 94% and 83% respectively (Table 6) and the CP performance of CWC(3, 1) and CWC(3, 5) with pt = Rabi season for p_i = 3 is 82% and 76% respectively (Table 7). In case of pt = Flowering (Tables 8 and 9), the CP values at p_i = 3 are significantly less than the results of Tables 6 and 7, as the duration of pt = Flowering is significantly less than the pt = year or season. However, from Tables 8 and 9 for p_i = 7, the CP values about 60% to 80% are realized for CWC(3, 1) and CWC(3, 5).

Given a sequence of periods of a given pt, the results show that for pts with large durations like year and season, the *reuse* can be achieved by preparing the suggestions for less number of periods. The results also

Table 8
CP values for pt = Flowering phenophase in Kharif season.

CWC (n, d)/period	CWC (1, 1)	CWC (2, 1)	CWC (3, 1)	CWC (4, 1)	CWC (5, 1)	CWC (1, 5)	CWC (2, 5)	CWC (3, 5)	CWC (4, 5)	CWC (5, 5)
3	82	78	**58**	42	38	94	62	**53**	40	22
5	100	90	**76**	58	40	98	82	**62**	44	30
7	100	94	**86**	76	61	98	90	**78**	62	44
10	100	98	**93**	82	78	100	94	**80**	68	44

Table 9
CP values for pt = Flowering phenophase in Rabi season.

CWC (n, d)/period	CWC (1, 1)	CWC (2, 1)	CWC (3, 1)	CWC (4, 1)	CWC (5, 1)	CWC (1, 5)	CWC (2, 5)	CWC (3, 5)	CWC (4, 5)	CWC (5, 5)
3	100	81	**61**	42	58	82	62	**22**	38	38
5	100	90	**78**	76	76	84	62	**54**	58	41
7	100	93	**82**	78	78	90	78	**61**	60	48
10	100	94	**84**	79	76	96	83	**70**	60	50

show that, for pts with small durations like phenophases of the Rice crop, the *reuse* can be achieved by preparing the suggestions for relatively more number of periods. Overall, for any domain including agriculture, the results provide an opportunity to improve the efficiency of weather-based agricultural DSSs by improving the *reuse* of the weather-based suggestions.

5. Summary and conclusion

Weather plays an important role in almost all aspects of the life. Weather-based DSSs are being built to improve the efficiency of the production systems in the domains of healthcare, agriculture, transport, governance and so on. In the literature, the notion of *reuse* is being exploited to improve the efficiency of DSS in several domains such as medical care and software project management. In this paper, we proposed a methodology to improve the *reuse* in weather-based DSSs. To improve *reuse*, the proposed methodology exploits weather category information available in the respective domain and proposes a notion of *Category-based Weather Condition (CWC)*. For the given domain, the domain experts divide the range of each weather variable into different categories based on its influence on the respective domain. The notion of CWC improves the *reuse*, as it improves the extent to find similar CWCs among the CWCs of different periods of the given period type.

We have analyzed the extent of *reuse* under the proposed framework by considering the case study of agromet advisory service operated by India Meteorological Department (IMD). We have conducted the experiments on 30 years of weather data collected at Rajendranagar, Hyderabad, Telangana State, India, by using the categories of weather variables provided by IMD and considering two types of CWCs with the duration equal to one day and five days. By varying the number of weather variables in CWC from one to five, we have computed the extent of *reuse* among CWCs of different periods by considering the period types: year, season, and phenophases (i.e., growth stages) of the Rice crop. The results show that there is a significant degree of *reuse* among the CWCs over years with respect to period types: year, season and phenophases of the Rice crop. Given a sequence of periods of a given period type, the results show that for period types with large durations like year and season, the *reuse* could be achieved by preparing the suggestions for less number of periods. The results also show that, for period types with small durations like phenophases of the Rice crop, the *reuse* can be achieved by preparing the suggestions for relatively more number of periods.

For any domain including agriculture, the results provide an opportunity to improve the efficiency of weather-based agricultural DSSs by improving the *reuse* of the weather-based suggestions. The proposed framework provides the following advantages for weather-based DSSs.

- Reduces the repetition of work: Normally, several domain experts are required to develop the content/suggestions for a given weather condition in the given domain. Under the proposed framework, once prepared, the content/suggestion can be reused for the similar weather condition. As a result, the proposed framework reduces the repetition of work by the domain experts.
- Reduce the operational cost: Under the proposed framework, several domain experts prepare the content/suggestions for a given weather condition as a one time activity. As a result, the operational cost of the system can be reduced as the system can be operated with less number of experts.
- Improves the quality: The proposed framework helps to improve the content quality as the eminent experts can be involved to prepare the high quality suggestions in a comprehensive manner for the given weather condition. Overall, the system improves the quality of weather based suggestions as it allows the *reuse* of high quality suggestions prepared by eminent experts for similar weather conditions.

As a part of the future work, we are planning to conduct extensive experiments on the weather data at multiple locations across India by considering the different period types and weather conditions with different durations. We are also planning to conduct the experiments by defining different types of weather conditions based on the distribution of weather values. Efforts will be made to explore the extent of *reuse* under the proposed framework for other domains such as healthcare, crops, livestock, transport and so on.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.compag.2018.12.010.

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